Posterior Summaries for Data Fusion

Working notes

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Scope

1 Posterior projection summaries for interpretable inference

A practical difficulty of Bayesian-nonparametric HTE models is that the posterior of $\tau(X)$ lives in an effectively infinite-dimensional space: each MCMC draw is a function, not a vector of coefficients. To produce inferentially valid summaries on an interpretable scale — coefficients on a small set of covariates, subgroup contrasts, tree-style rules — we adopt the posterior projection approach of Woody et al. (2020). The idea is to map each posterior draw into a simpler function class via a deterministic projection and to treat the collection of projected summaries as the posterior over the summary. No model is refit; uncertainty is inherited exactly.

1.1 The projection

Let $\mathcal{G} = \{g_\gamma(x) = \phi(x)^\top \gamma : \gamma \in \mathbb{R}^k\}$ be a target function class indexed by a coefficient vector γ, with basis $\phi : \mathcal{X} \to \mathbb{R}^k$. Typical choices for ϕ include main effects of a chosen subset of covariates, low-order interactions, B-spline bases for continuous covariates, or indicator dummies for a small set of subgroups.

Given a posterior draw of the target function $\theta^{(r)}(\cdot)$, where θ stands for any of τ, c, $\Delta_{\mathrm{RMST}}(t^*, \cdot)$, or $\Delta_{\mathrm{RD}}(t, \cdot)$, the projection at iteration r is the solution to the weighted least squares problem

$$\gamma^{(r)} = \arg\min_{\gamma \in \mathbb{R}^k} \sum_{i=1}^{n_{\mathrm{ev}}} w_i^{(r)} \big(\theta^{(r)}(x_i) - \phi(x_i)^\top \gamma\big)^2 = (\Phi^\top W^{(r)} \Phi)^{-1} \Phi^\top W^{(r)} \theta^{(r)}, \tag{1}$$

where Φ is the $n_{\mathrm{ev}} \times k$ design matrix with rows $\phi(x_i)^\top$, $\theta^{(r)} = (\theta^{(r)}(x_1), \ldots, \theta^{(r)}(x_{n_{\mathrm{ev}}}))^\top$, and $W^{(r)} = \mathrm{diag}(w_1^{(r)}, \ldots, w_{n_{\mathrm{ev}}}^{(r)})$ is an optional weight matrix used either to focus on a target population or to inject covariate-distribution uncertainty via a Bayesian bootstrap.

The collection $\{\gamma^{(r)}\}_{r=1}^R$ is the posterior over the projection. Posterior means, marginal credible intervals, and probabilities of the form $\Pr(\gamma_k > 0 \mid \text{data})$ are computed as empirical operations on this collection.

1.2 Validity

Because $\gamma^{(r)}$ is a deterministic function of the posterior draw $\theta^{(r)}$ (and the bootstrap weights $w^{(r)}$, when used), the collection $\{\gamma^{(r)}\}$ is a sample from the posterior distribution of the pushforward of θ under the projection map. No additional modelling assumption is introduced. The credible intervals derived from $\{\gamma^{(r)}\}$ inherit the full uncertainty of the original posterior of θ, in contrast to the common workflow of fitting an interpretable model to a point summary (e.g. the posterior mean) and reporting that model's standard errors, which double-uses the data and ignores the original posterior variance.

1.3 What to project, and on which scale

In our setting three projections are natural:

(P1) The CATE on the AFT scale: $\tau(X)$. The cleanest interpretation: γ_k is the log-time shift per unit of covariate k. Useful for statistical reporting and for diagnostics of the structure of heterogeneity.

(P2) The survival-scale contrast: $\Delta_{\mathrm{RMST}}(t^*, X)$. For clinically meaningful t^* (e.g. in-trial and well past trial follow-up), the projection coefficients have the interpretation of "RMST gain per unit of x at horizon t^*". This is the headline summary for the audience interested in the long-term goal (G2). Reporting projections at multiple t^* traces how the heterogeneity summary moves into the extrapolation regime.

(P3) The confounding function: $c(X)$. A diagnostic projection rather than a clinical summary. The projection of $c(X)$ describes which covariates carry the residual bias signal in the OS. If c admits a clean low-dimensional projection, the bias structure is interpretable; if it does not, the OS contribution is intrinsically complex and the reader should be warned.

1.4 The headline use: RCT-only vs. fusion contrast

The single most informative deployment of the projection in this paper is to fit the model under two data configurations — RCT-only and full RCT+OS fusion — and to project both posteriors onto the *same* basis ϕ at the *same* evaluation points. The marginal posteriors of each γ_k can then be plotted side by side, giving a direct, scale-matched answer to the question "how did borrowing change the heterogeneity story?". The same construction with the model fit once and the evaluation points changed (RCT-population vs. OS-population empirical distribution) decomposes "model effect" from "population effect" in the apparent change of heterogeneity.

1.5 Bayesian-bootstrap variant

When the summary target is a functional of the covariate distribution (e.g. a marginal effect, or any quantity defined by averaging over the empirical distribution), the posterior of the model parameters alone does not capture sampling uncertainty in that distribution. Following Rubin (1981) and Antonelli et al. (2019), we inject covariate-distribution uncertainty by drawing Dirichlet weights at each MCMC iteration:

$$w^{(r)} = (w_1^{(r)}, \ldots, w_{n_{\mathrm{ev}}}^{(r)}) \sim \mathrm{Dirichlet}(1, 1, \ldots, 1), \tag{2}$$

and use these as the weights in (1). The same Dirichlet draw is used everywhere it appears in the iteration (e.g. when also computing a marginal RMST as a weighted average over evaluation points), so that the dependence is preserved.

1.6 Algorithm

Algorithm 1 Posterior projection summary

Require: posterior draws of model parameters $\{\theta^{(r)}_{\text{model}}\}_{r=1}^{R}$; target functional $\theta(\cdot)$ (one of τ, c, $\Delta_{\text{RMST}}(t^*, \cdot)$, etc.); basis $\phi : \mathcal{X} \to \mathbb{R}^k$; evaluation points $\{x_i\}_{i=1}^{n_{\text{ev}}}$; flag `use_bb` for Bayesian bootstrap.

Assemble the design matrix $\Phi \in \mathbb{R}^{n_{\text{ev}} \times k}$ with rows $\phi(x_i)^\top$; centre and (optionally) scale columns of Φ.
Precompute the QR decomposition of Φ if `use_bb` is false; otherwise the linear system is re-solved at each iteration.
for $r = 1, \ldots, R$ **do**
    Evaluate the target at all evaluation points: $\theta^{(r)}_i \leftarrow \theta(x_i; \theta^{(r)}_{\text{model}})$ for $i = 1, \ldots, n_{\text{ev}}$.
    if `use_bb` **then**
        Draw $w^{(r)} \sim \text{Dirichlet}(1, \ldots, 1)$.
        $\gamma^{(r)} \leftarrow (\Phi^\top W^{(r)} \Phi)^{-1} \Phi^\top W^{(r)} \theta^{(r)}$.
    else
        $\gamma^{(r)} \leftarrow \Phi^+ \theta^{(r)}$ via the precomputed QR factorisation, where Φ^+ denotes the Moore–Penrose pseudoinverse.
    end if
    Store $\gamma^{(r)}$.
end for
return $\{\gamma^{(r)}\}_{r=1}^{R}$.

Posterior summaries are then computed in the obvious way:

$$\bar{\gamma}_k = \frac{1}{R} \sum_{r=1}^{R} \gamma_k^{(r)}, \qquad \widehat{\text{CrI}}_\alpha(\gamma_k) = \left[Q_{\alpha/2}(\gamma_k^{(r)}),\ Q_{1-\alpha/2}(\gamma_k^{(r)})\right], \qquad \widehat{\Pr}(\gamma_k > 0) = \frac{1}{R} \sum_{r=1}^{R} \mathbf{1}\{\gamma_k^{(r)} > 0\},$$

with joint summaries obtained from the empirical joint distribution of $\{\gamma^{(r)}\}$ (e.g. posterior probability that γ_j and γ_k have the same sign, or that a chosen contrast exceeds a threshold).

1.7 Implementation notes

Basis selection. The basis ϕ should be small and fixed: five to ten terms is typically enough for an interpretable summary, and a fixed basis is essential for validity of the projection-posterior construction. If basis selection is desired, do it once on a point summary of the posterior (e.g. run a lasso of the posterior mean of $\theta(\cdot)$ on a candidate dictionary), and then fix the selected basis for the projection step. Repeating selection at every MCMC iteration would introduce non-deterministic dependence on the draws and break the clean "pushforward" interpretation of Woody et al. (2020); this is feasible but adds complexity that we do not pursue.

Centring and scaling. Centre and scale columns of Φ before the projection so that coefficient magnitudes are comparable and the intercept absorbs the mean of $\theta^{(r)}$. Standard regression hygiene; matters more once relative magnitudes of γ_k are interpreted visually.

Evaluation set. The choice of evaluation set determines the implicit target population of the summary. Pooled RCT+OS rows describe the combined sample; OS rows alone describe the target observational population (the natural choice under (G1)); RCT rows describe the trial population (typically the natural choice for HTA-style extrapolation under (G2)). Always state explicitly which set is used.

Tree projection (CART summaries). The procedure generalises to projection onto a regression tree by replacing the OLS step with a tree fit. The complication is that the tree *structure* can change across MCMC draws, which makes summarisation awkward. Two clean options: (i) fix the tree structure (e.g. the tree fit to the posterior mean of θ) and refit only the leaf values at each iteration — this stays within the Woody–Carvalho–Murray framework; (ii) report subgroup membership probabilities, summarising the variability of the partition itself. Option (i) is simpler and is recommended unless the partition itself is the object of interest.

Choice of t^* for the RMST projection. If $t^* \leq t_{\mathrm{RCT}}$ the projection summary reflects RCT-anchored information; if $t^* > t_{\mathrm{RCT}}$ the summary is computed from the extrapolated posterior and depends on the HDP and the OS. Reporting projections at both an in-window and an out-of-window t^* traces how the heterogeneity story evolves into the extrapolation regime, complementing the coverage-vs-horizon diagnostic from the simulation.

Caveat on interpretation. The projection is a *summary*, not the truth. If $\tau(X)$ is genuinely nonlinear in covariates, a linear projection captures the best linear approximation in MSE; the nonlinear structure is absorbed into residual variation and not reported. This is the price of interpretability and should be acknowledged when the summary is presented.

Computational cost. Negligible. Each iteration costs one k-dimensional linear solve on $\Phi^\top\Phi$ (or a single back-substitution with a cached QR), plus the cost of evaluating $\theta^{(r)}(x_i)$ for each i. The latter dominates only when $\theta = \Delta_{\mathrm{RMST}}$, which requires a numerical integration per evaluation point; in that case the integration can be cached at fixed quadrature nodes across iterations.

2 Related work

The reframing places the project at the intersection of three literatures that have historically not communicated very much: causal data fusion, Bayesian survival extrapolation for HTA, and flexible Bayesian survival modelling. We sketch each below and indicate where the proposed method sits.

2.1 Survival extrapolation for HTA

The foundational reference is NICE DSU Technical Support Document 14 (Latimer, 2013), which codified the practice of fitting standard parametric AFT/PH families (exponential, Weibull, log-logistic, lognormal, Gompertz, generalised gamma) to trial data and selecting among them by fit and clinical plausibility. The TSD 14 process is widely followed but also widely criticised: fits in the observed window are nearly indistinguishable, while extrapolations diverge dramatically (Bell Gorrod et al., 2019; Bagust and Beale, 2014).

External data have been used in HTA extrapolation in several distinct ways:

- **General-population mortality** as a lower bound on the hazard via additive-hazards (relative survival) models;
- **Registry conditional survival** from cancer registry databases used as external survival information for cancer (Guyot et al., 2017);
- **Expert elicitation** treated as prior data on long-term hazard (Guyot et al., 2017);
- **Longer-follow-up surrogate trials** such as the cilta-cel/LEGEND-2 example (Vyas et al., 2023), in which a phase I study with longer follow-up informs extrapolation of a pivotal phase II trial through informative priors on shape parameters.

The most directly comparable methodological piece is Jackson (2023)'s `survextrap`, which fits an M-spline hazard with a hierarchical prior on the coefficients, jointly to individual-level RCT data and external (possibly aggregate) data using Bayesian multi-parameter evidence synthesis. `survextrap` supports proportional and flexible non-proportional hazards, waning treatment effects, cure models, and additive-hazards formulations (Jackson, 2023). Empirical evaluations (Timmins et al., 2025) report that the model gives accurate long-term extrapolations when long-term external data on the patient population of interest are included, and quantifies structural uncertainty appropriately when they are not.

Where the proposed method differs from `survextrap`. `survextrap` does not address unmeasured confounding in the external data; the external sources it accommodates are general-population mortality, registry conditional survival, and elicited beliefs — not a confounded observational study of the *same* treatment. The proposed framework adds an explicit confounding-function correction $c(X)$, allowing the external data to be a contemporaneous OS subject to non-randomised treatment assignment, not merely a background hazard reference. The modelling style also differs: M-spline hazards versus BART forests on AFT components, and a flexible parametric error versus an HDP nonparametric error.

2.2 Causal data fusion under unmeasured confounding

The methodological core — a confounding function $c(X)$ correcting OS contrasts that do not hold unconfoundedness — is established in Yang et al. (2025), who derive semiparametric efficiency bounds for joint estimation of τ and c. Kallus et al. (2018) propose the "experimental grounding" framework, in which an RCT supplies a bias function correcting an observational HTE estimator. Yao et al. (2024) extend data fusion to partial OS treatment information. van der Laan et al. (2024) develop adaptive TMLE for RCT–RWD fusion that adapts borrowing to detected bias.

The closest match to the proposed work is Ye et al. (2025), who develop an integrative AFT framework for high-dimensional RCT and RWD with censoring and hidden confounding, using Stute-weighted penalised regression with separate regularisation for τ and the confounding component. The proposed method is essentially a Bayesian-nonparametric counterpart to Ye et al. (2025): BART ensembles in place of penalised parametric models, nonparametric error in place of a Gaussian or assumed AFT family, and commensurate-prior structure on the baseline split between sources.

A practical caveat from Lin et al. (2024): bias-correction methods for marginal ATE estimation in this literature have not delivered RMSE improvements over RCT-only analysis in their simulations, and the methods' theoretical advantage is for the *heterogeneous* effect function rather than its average. The proposed framing targets HTE plus long-term extrapolation, both of which are tasks for which RCT-only inference is genuinely deficient — the latter, simply because it cannot be done from the trial alone.

2.3 Bayesian flexible survival modelling

The BART-based modelling of μ, g, τ, c rests on the Bayesian causal forest (BCF) line of work (Hahn et al., 2020), which introduced the prognostic/treatment-effect decomposition with separate prior scales that the four-forest version generalises. The HDP error builds on the centred stick-breaking mixture of Yang et al. (2010) and the wider Bayesian-nonparametric survival literature; the hierarchical extension specifically lets the source label index the group hierarchy, with shared atoms across sources.

2.4 Positioning summary

	Confounding correction	Long-term extrapolation	Nonparametric error
Latimer (2013) (TSD 14 workflow)	—	yes	—
Guyot et al. (2017)	—	yes	—
Jackson (2023) (`survextrap`)	—	yes	flexible
Ye et al. (2025)	yes	limited†	—
Yang et al. (2025)	yes	—	—
Kallus et al. (2018), Yao et al. (2024)	yes	—	—
This work	**yes**	**yes**	**yes (HDP)**

Table 1: Schematic positioning. †: Ye et al. (2025) targets HTE estimation in the joint AFT framework; long-term extrapolation is not its stated goal but is in principle compatible with the AFT structure.

The niche is clear: the HTA-pragmatic literature handles extrapolation flexibly but does not correct for confounding in the external source; the causal-fusion literature corrects for confounding but does not target extrapolation; and few methods combine the two with a fully nonparametric error distribution. The proposed method sits deliberately in that gap.

References

Antonelli, J., Parmigiani, G., and Dominici, F. (2019). High-dimensional confounding adjustment using continuous spike and slab priors. *Bayesian Analysis*, 14(3):805–828.

Bagust, A. and Beale, S. (2014). Survival analysis and extrapolation modeling of time-to-event clinical trial data for economic evaluation: an alternative approach. *Medical Decision Making*, 34(3):343–351.

Bell Gorrod, H., Kearns, B., Stevens, J., Thokala, P., Labeit, A., Latimer, N., Tyas, D., and Sowdani, A. (2019). A review of survival analysis methods used in NICE technology appraisals of cancer treatments: consistency, limitations, and areas for improvement. *Medical Decision Making*, 39(8):899–909.

Guyot, P., Ades, A. E., Ouwens, M. J. N. M., and Welton, N. J. (2012). Enhanced secondary analysis of survival data: reconstructing the data from published Kaplan–Meier survival curves. *BMC Medical Research Methodology*, 12:9.

Guyot, P., Ades, A. E., Beasley, M., Lueza, B., Pignon, J.-P., and Welton, N. J. (2017). Extrapolation of survival curves from cancer trials using external information. *Medical Decision Making*, 37(4):353–366.

Hahn, P. R., Murray, J. S., and Carvalho, C. M. (2020). Bayesian regression tree models for causal inference: regularization, confounding, and heterogeneous effects. *Bayesian Analysis*, 15(3):965–1056.

Jackson, C. H. (2023). survextrap: a package for flexible and transparent survival extrapolation. *BMC Medical Research Methodology*, 23:282.

Kallus, N., Puli, A. M., and Shalit, U. (2018). Removing hidden confounding by experimental grounding. In *Advances in Neural Information Processing Systems*, 31.

Latimer, N. R. (2013). Survival analysis for economic evaluations alongside clinical trials—extrapolation with patient-level data: inconsistencies, limitations, and a practical guide. *Medical Decision Making*, 33(6):743–754. (NICE DSU Technical Support Document 14.)

Lin, X., Tarp, J. M., and Evans, R. J. (2024). Data fusion for efficiency gain in ATE estimation: a practical review with simulations. *arXiv preprint* arXiv:2407.01186.

Rubin, D. B. (1981). The Bayesian bootstrap. *The Annals of Statistics*, 9(1):130–134.

Timmins, I. R., Jackson, C. H., and colleagues (2025). A simulation and case study to evaluate the extrapolation performance of flexible Bayesian survival models when incorporating real-world data. *arXiv preprint* arXiv:2505.16835.

van der Laan, M. J., Qiu, S., and van der Laan, L. (2024). Adaptive-TMLE for the average treatment effect based on randomized controlled trial augmented with real-world data. *arXiv preprint* arXiv:2405.07186.

Vyas, M., Hettle, R., Rana, M., Stevenson, K., et al. (2023). Extrapolation of survival data using a Bayesian approach: a case study leveraging external data from cilta-cel therapy in multiple myeloma. *Advances in Therapy*, 40(11):4877–4893.

Woody, S., Carvalho, C. M., and Murray, J. S. (2020). Bayesian inference with posterior projection. *Journal of Computational and Graphical Statistics*, 29(4):798–808.

Yang, M., Dunson, D. B., and Baird, D. (2010). Semiparametric Bayes hierarchical models with mean and variance constraints. *Computational Statistics & Data Analysis*, 54(9):2172–2186.

Yang, S., Liu, S., Zeng, D., and Wang, X. (2025). Data fusion methods for the heterogeneity of treatment effect and confounding function. *Bernoulli*, 31(4):2987–3012.

Yao, D., Tang, C., Cui, Q., and Li, L. (2024). Combining incomplete observational and randomized data for heterogeneous treatment effects. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*.

Ye, X., Yang, S., Wang, X., et al. (2025). Integrative analysis of high-dimensional RCT and RWD subject to censoring and hidden confounding. *Lifetime Data Analysis*, 31:473–497.